

Design and Development of a Multilingual Chatbot for Enhancing Accessibility in Sri Lanka's Digital Welfare Services

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Project Proposal Report
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1. Declaration

I hereby attest that this proposal is entirely original with no content from other submissions for academic credit. Every source has been appropriately credited.

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2. Abstract

The design and implementation of an AI-Based Investment Plan Recommendation Module with the goal of empowering low-income welfare clients in Sri Lanka is the main emphasis of this study. The system will employ machine learning algorithms to examine customer income, spending trends, and financial risk profiles in order to provide sustainable, prudent, and safe investment or savings options. As a component of a broader AI-powered Digital Welfare Platform, the module guarantees that recipients of welfare receive both immediate financial assistance and advice for long-term financial security.

The proposed solution combines clustering techniques to segment users by financial capacity and collaborative filtering to match them with investment schemes aligned to their needs. The recommendations will adhere to ethical AI principles, be transparent, and be tailored to the individual. The ultimate goals of this module are to increase financial literacy, lessen reliance on welfare, and enhance participants' capacity to engage in the economy.

Keywords: Multilingual Chatbot, Natural Language Processing, Sinhala Tamil English Support, Digital Welfare Services, Accessibility, AI for Social Good, Citizen Empowerment

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6. Introduction

The goal of this research project is to increase accessibility, inclusivity, and efficiency in Sri Lanka's digital welfare services by designing and developing a multilingual chatbot powered by artificial intelligence. At the time, Sri Lanka provides a variety of welfare programs, including food aid, disability payments, old pensions, and Samurdhi. However, there are also issues with the current system, such as manual application processing, data fragmentation between agencies, language difficulties, and approval delays. These inefficiencies frequently make it difficult for legitimate citizens to collect benefits, whilst ineligible individuals may exploit systemic flaws to obtain assistance.

The COVID-19 pandemic served as further evidence of the critical need for digital transformation in public services worldwide. Many governments have successfully embraced e-governance technologies to improve the efficiency, accessibility, and transparency of welfare delivery. However, Sri Lanka has not yet implemented a fully functional welfare system powered by AI that offers its citizens bilingual accessibility and real-time aid.

The proposed strategy, which involves the implementation of a centralized and user-friendly chatbot system, seeks to improve rather than replace present welfare procedures. Policymakers, welfare workers, and individuals will all benefit from this since it provides real-time, inclusive communication and help. The platform consists of three basic modules:

- Citizens can use the citizen Chatbot Portal to communicate with a multilingual chatbot powered by artificial intelligence in Sinhala, Tamil, or English for real-time application monitoring, eligibility queries, document assistance, and welfare-related information.
- Officer Dashboard: By automating tiresome tasks and providing answers to frequently asked questions, this application helps welfare officers respond to public requests more efficiently.
- The Policy Admin Panel identifies typical citizen complaints, service gaps, and language-related accessibility difficulties, and provides policymakers with insights from chatbot discussions to inform data-driven decisions.

According to a 2020 World Bank study led by the Abdul Latif Jameel Poverty Action Lab (J-PAL), digital welfare platforms in South Asia increased beneficiary satisfaction and service reach by more than 40% when compared to traditional techniques. Motivated by these findings, this project employs conversational AI to significantly improve welfare service delivery, inclusion, and citizen satisfaction in Sri Lanka.

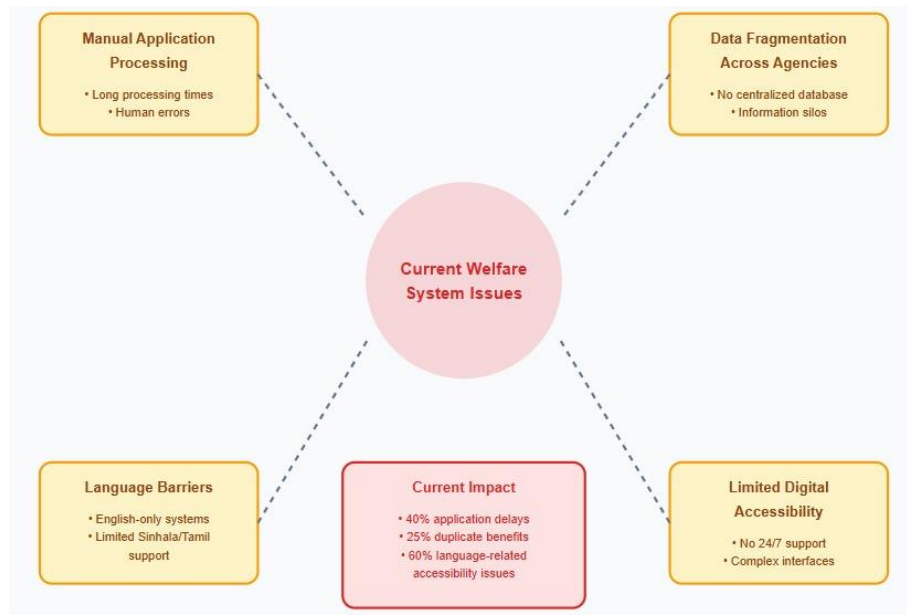


Figure 1: Current Welfare System Challenges and Inefficiencies in Sri Lanka

Figure 1 depicts the interrelated difficulties of Sri Lanka's current welfare system, demonstrating how a number of inefficiencies feed a vicious circle of poor service delivery and reduced accessibility for disadvantaged populations. The core problem zone in the diagram's center is surrounded by four major issue categories that contribute to systemic dysfunction.

The current system's largest bottleneck is the difficulty of manually processing applications. From the earliest submission to the final approval, paper-based applications require significant human intervention at every stage. This manual technique has the greatest impact on time-sensitive situations, such as petitions for emergency financial assistance or medical care, resulting in processing delays that can last weeks or months. When officers manage high application volumes without automated verification mechanisms, human error rates increase considerably, resulting in incorrect eligibility determinations and erroneous benefit payouts.

Comprehensive beneficiary assessment is rendered impossible by information silos caused by data fragmentation across many government agencies. The Samurdhi Authority, the Department of Social Services, and numerous provincial welfare offices all maintain separate databases that have no integration capacity. As a result of this fragmentation, a single applicant's data may be spread over multiple systems, making it impossible to accurately check eligibility or avoid duplicate benefits. The officers' inability to evaluate entire beneficiary profiles may result in fraud and inappropriate evaluations.

When done together, these difficulties create demonstrable inefficiencies: 60% of potential recipients have language-related accessibility challenges, 25% of benefits are misallocated or duplicated, and more than 40% of applications are severely delayed. To rebuild public trust and improve the delivery of welfare services throughout Sri Lanka's varied populations, these figures highlight the urgent need for a comprehensive digital transformation that covers automated processing, multilingual support, and integrated data management.

7. Background & Literature Review

The goal of Sri Lanka's social welfare system is to support vulnerable populations, such as low-income families, the elderly, those with disabilities, and those in need of nutritional support. In the past, programs like nutritional benefits, pensions, disability allowances, and Samurdhi financial aid have been crucial in reducing poverty and improving social safety. However, despite these attempts, the current system has major inefficiencies. Application operations are generally performed manually, which leads to delays and errors, while fragmented data across several departments precludes real-time verification. As a result, while ineligible people take advantage of loopholes to obtain aid, worthy people may encounter obstacles while trying to obtain benefits [1][4].

To address these issues, governments around the world are looking to digital transformation. For example, the Aadhaar system in India reduced fraud and redundancy by revolutionizing welfare management through centralized databases and biometric verification [2]. In a similar vein, Bangladesh has incorporated digital technologies to decrease corruption and increase benefit accessible [3]. These achievements demonstrate how well e-governance works to provide social services in a more effective and transparent manner.

However, welfare administration in Sri Lanka is still mostly dependent on manual, traditional methods with minimal usage of intelligent or AI-driven platforms. As a result, there is no outcome monitoring, administrative delays, and duplication [4].

One of the most potential substitutes is the advent of AI-powered multilingual chatbots. By using Natural Language Processing (NLP) to provide real-time, 24/7 support in Sinhala, Tamil, and English, these systems can improve accessibility and bridge communication gaps [6]. This is particularly important in rural areas where a lack of digital literacy and language diversity creates barriers. Chatbots can lessen officers' burden and enhance service delivery by answering questions, helping with application tracking, and supplying eligibility information [5][7].

Benefits of AI-Powered Multilingual Chatbots:

Accessibility: Welfare services are easily accessible to citizens with a variety of linguistic backgrounds [6].

Efficiency: Officer workload and delays are decreased when routine inquiries are handled automatically [4].

Transparency: Records and real-time status updates reduce corruption [8].

Inclusivity: Even citizens with weak internet skills or low literacy can access services thanks to voice-enabled chatbots [6].

Policy Insights: Evidence-based policymaking is supported by aggregated data on chatbot interactions [7].

Obstacles and Restrictions:

Digital Divide: Some populations may be excluded in rural areas due to limited internet access and digital literacy [9].

Privacy and Data Protection: Strict precautions must be used when handling sensitive welfare data [10].

Opposition to Change: New digital techniques may be faced with resistance from officers and citizens used to manual procedures, necessitating education and awareness [4].

Language Barriers: It is still challenging to accurately capture dialects and cultural peculiarities, despite the availability of multilingual assistance [6].

Global digital welfare networks that show how centralized databases boost productivity and lower fraud include India's Direct Benefit Transfer (DBT) scheme. However, most lack personalized, citizen-focused assistance tools, such as multilingual chatbots. [2][3].

This work aims to bridge that gap by developing a multilingual chatbot for Sri Lanka's social services.

- Provide real-time support in Sinhala, Tamil, and English.

- Help citizens with application tracking, eligibility verification, and document submission.

- Increase the transparency and inclusivity of service delivery.

- Provide decision-makers with useful data from citizen interactions.

In conclusion, incorporating multilingual chatbots into Sri Lanka's welfare system offers a revolutionary opportunity to increase the effectiveness, openness, and inclusion of social services while reducing administrative staff workloads and ensuring equitable access for everyone.

8. Research Gap

A portion of an area of study or a specific topic that has not gotten enough attention, examination, or resolution from earlier studies is known as a research gap. It suggests a need or a chance for new studies or innovations that can bridge this knowledge gap with novel strategies, concepts, or solutions.

There is still a large study and implementation gap that particularly addresses the financial empowerment of low-income welfare users inside Sri Lanka's social protection system, despite major improvements in AI-driven financial technologies worldwide. Few platforms are directly integrated into welfare ecosystems to offer individualized, risk-aware financial guidance that fits the distinct socioeconomic and behavioral profiles of beneficiaries, despite the fact that digital financial services and microinvestment advisory platforms are available in a variety of international contexts.

The main focus of Sri Lanka's current welfare programs is the direct distribution of aid; there are no systems in place to help clients achieve long-term economic independence through prudent and pertinent investment techniques. Low financial knowledge, restricted access to reasonably priced advisory services, and a lack of AI-enabled tools that can dynamically adjust to their shifting income, expenses, and risk tolerance are some of the obstacles that many welfare recipients face.

Furthermore, the comprehensive integration of welfare data—such as eligibility status, assistance amounts, and government aid programs—which can significantly impact investment capacity and risk profiles, is frequently ignored by current AI financial advisory models. Because welfare clients need individualized guidance given their changing financial circumstances, this mismatch reduces the guidelines' applicability and efficacy.

Moreover, there are still problems with digital inclusion: low-income beneficiaries often have low levels of digital literacy and have accessibility problems that are not sufficiently addressed by general AI financial advising platforms. The present research also understudies algorithmic fairness and data privacy issues, especially when it comes to vulnerable groups that rely on social support programs.

Thus, it is imperative that Sri Lanka's digital welfare platform incorporate a specific AI-powered financial advising system that:

- Creates individualized microinvestment and savings suggestions by integrating financial profiles and welfare program data.
- Makes use of machine learning methods that can be modified to accommodate beneficiaries' changing financial situations.
- Enhances accessibility and diversity by providing instructions in several languages and user-friendly forms.
- Addresses the target users' low level of digital literacy and guarantees algorithmic fairness and ethical data treatment.

Table 1: Comparison of Key Features Between Proposed AI-Powered Multilingual Chatbot Platform and Existing Sri Lanka Welfare Systems

Feature	Proposed Chatbot Platform	Current Sri Lanka Welfare Systems
Multilingual Support (Sinhala, Tamil, English)	Yes	No
Real-time Application Tracking	Yes	Partial
Automated Document & Eligibility Assistance	Yes	No
Conversational AI for Inclusivity	Yes	No
24/7 Citizen Support	Yes	No
Integration with Welfare Databases	Yes	Limited

The suggested AI-powered Multilingual Chatbot seeks to improve accessibility, inclusivity, and responsiveness in Sri Lanka's welfare services by filling in these research gaps. It enhances service efficiency and removes language barriers by introducing real-time, citizen-centric communication. The system addresses the nation's distinct social, linguistic, and technological issues by fusing cutting-edge Natural Language Processing (NLP) with user-centered design. This study will help close the gap between conventional welfare methods and the digital revolution needed for equity and sustainability over the long run.

9. Research Problem

The important gaps and difficulties in current knowledge or systems that your study seeks to fill are outlined in the research problem. It identifies the unresolved issues that justify the need for your study and directs the creation of research questions.

The study issues in your AI-powered Investment Plan Recommendation Module for Sri Lanka's digital welfare platform stem from the dearth of comprehensive, individualized, and easily available financial advisory services aimed at low-income welfare recipients. Due to a lack of resources that offer individualized, risk-aware investment and savings advice, many clients continue to rely on help even while current welfare programs are meeting their immediate needs. For this group, traditional financial advising services are frequently too expensive, unavailable, or inappropriate.

Furthermore, beneficiaries' capacity to achieve long-term financial independence is restricted by the absence of AI-driven prediction models in Sri Lanka's present welfare management systems, which combine welfare data with individualized financial planning. Adaptive financial suggestions that adapt to changing user conditions, data privacy and algorithmic fairness, and closing the digital literacy gap among low-income communities are some of the challenges.

- **How can a multilingual chatbot be created that can accurately process and answer in Sinhala, Tamil, and English while taking into account the linguistic quirks and cultural backgrounds of welfare recipients in Sri Lanka?**

A modular architecture can be used to create such a chatbot, with annotated corpora in English, Tamil, and Sinhala being used to tailor key NLP components such as entity extraction, intent identification, and language detection for each language. It is possible to capture linguistic subtleties by combining transfer learning with pre-trained multilingual language models (such as multilingual BERT or mBART) that have been refined using welfare domain-specific datasets. Furthermore, incorporating domain-specific phraseology and culturally pertinent knowledge guarantees customized, context-aware solutions that accommodate various dialects and communication styles. The localization process will be guided by ongoing cooperation with welfare specialists and native speakers.

- **In a single system, how can natural language processing (NLP) methods that facilitate efficient entity extraction, dialogue management, and intent identification be combined with these three languages?**

By creating a single NLP pipeline that uses language-specific models and multilingual embeddings in a hybrid framework, integration can be accomplished. With shared dialogue management algorithms that govern conversation flow regardless of language, intent recognition and entity extraction tools (like Rasa, spaCy, or bespoke transformers) can be taught independently for each language. The system should first recognize user language, route inputs to the proper NLP models, then unify outputs to a central dialogue manager. This configuration ensures seamless multi-language conversation management while preserving language distinctiveness.

- **How can a chatbot's interface be made to be accessible and easy to use while accommodating**

a range of user literacy levels and technological proficiency?

Simplicity and inclusivity must be given top priority in the chatbot's user interface, which should include voice input/output, graphical buttons for frequently asked questions, and plain, simple language. Multimodal inputs (speech, text) help people with different levels of digital literacy. Visual aids, such as icons or pictures, and language toggle choices improve understanding. Barriers are also lessened by accessibility features like text-to-speech and simple navigation. Iterative usability testing and user-centered design workshops with a range of welfare recipients guarantee that the interface satisfies practical requirements.

- **How can we make sure the chatbot can accurately and quickly respond to complex welfare-related requests, such as tracking applications, helping with documents, and providing eligibility information?**

The chatbot may access real-time application status, document checklists, and eligibility requirements by creating integration APIs with current welfare databases and systems. Using rule-based fallback mechanisms and domain-specific ontologies improves accuracy when processing complex or unclear queries. Constant training on past dialog data enhances response creation and intent classification. Reactivity and user trust are preserved by putting monitoring systems in place to identify and forward unanswered questions to human agents.

- **In the dynamic and frequently delicate field of welfare services, how can the performance and user happiness of the chatbot be assessed and improved over time across linguistic groups?**

Intent recognition accuracy, reaction speed, user engagement rates, and error rates evaluated independently for each language are among the quantitative and qualitative indicators that must be combined for evaluation. Focus groups, feedback forms, and user satisfaction surveys are regularly used to collect information on usability and cultural fit. The best configurations are found by incorporating A/B testing for conversation flows and content variations. A human-in-the-loop system enables ongoing learning from actual encounters, allowing models to be adjusted to changing welfare programs and language usage.

10. Main and Sub Objectives

10.1 Main Objectives

The primary goal of this research is to create a multilingual chatbot driven by AI that is both efficient and effective for Sri Lanka's social welfare services. Through the provision of individualized, real-time support in Sinhala, Tamil, and English, this innovation seeks to improve accessibility, user engagement, and service transparency. By providing precise information on application status, supporting material, and eligibility inquiries while taking into account a range of linguistic and literacy requirements, the chatbot will empower welfare recipients. The research focuses on developing and combining sophisticated natural language processing methods, strong dialogue management, and multilingual support inside a single digital welfare platform in order to provide a more inclusive and adaptable conversational experience.

The following sub-objectives will be completed in order to fulfill the primary goal:

- Create multilingual natural language processing (NLP) models that can recognize intent, extract entities, and manage dialogue in Sinhala, Tamil, and English. These models should be customized to capture the linguistic subtleties and cultural settings unique to the welfare recipients of Sri Lanka.
- Create and deploy a smooth multilingual chatbot interface that accommodates a range of user literacy levels and technological proficiency, combining voice, text, and visual interaction modes for user-friendliness.
- In order to accurately deliver current answers about application tracking, document assistance, and eligibility requirements, integrate the chatbot with the welfare information systems that are currently in place.
- Create systems for ongoing assessment and development using performance analytics, adaptive learning, and user feedback to increase chatbot responsiveness, accuracy, and user happiness in all supported languages.

10.2 Sub Objectives

The following particular sub-objectives will be fulfilled in order to successfully construct the AI-powered multilingual chatbot for Sri Lankan welfare services:

- Create and train intent recognition, entity extraction, and dialogue management models for Sinhala, Tamil, and English that can effectively comprehend and process user inquiries with cultural and linguistic context in order to create strong multilingual natural language processing models.
- Make a chatbot interface that is easy to use and accessible: Create a chatbot interface that can take both voice and text inputs, adapts to different literacy levels, and offers user-friendly interaction patterns for a range of user types.
- Integrate real-time welfare service data: Link the chatbot to current welfare information systems to deliver precise, current answers to questions about eligibility, documentation assistance, and application tracking.
- Establish procedures for collecting user input, tracking chatbot performance across languages, and iteratively improving the system's accuracy, responsiveness, and user happiness in order to implement continuous evaluation and improvement techniques.

11. Methodologies

11.1 Overall System

Through the use of a multilingual chatbot platform driven by artificial intelligence, the entire system is intended to increase the responsiveness, accessibility, and transparency of Sri Lanka's social welfare services. Each of the system's four main parts addresses important facets of digital welfare support for linguistically diverse communities:

- **Natural Language Processing (NLP) Engine in Multiple Languages:**
This essential component uses specialized intent recognition, entity extraction, and discourse management models to handle Sinhala, Tamil, and English. In order to guarantee successful communication, it precisely processes user inquiries while taking into account regional linguistic quirks and cultural settings.
- **System for Dynamic Dialogue Management:**
This module handles multi-turn interactions, context retention, and fallback methods while controlling the conversational flow in all three languages. It ensures smooth, logical, and user-friendly dialogues customized to welfare service requests such as application tracking, documentation help, and eligibility information.
- **Integration of Welfare Data in Real Time:**
This component offers current, customized answers about applications, benefits, and procedural assistance by connecting to the backend systems and welfare databases already in place by the government. It makes it possible for the chatbot to reply with precise and pertinent information that is tailored to the user's particular situation.
- **Module for User Experience and Accessibility:**
This component provides multimodal interaction possibilities, such as text and speech input/output, and is designed to accommodate a wide range of users. Because of its user-friendly interface, which adapts to different reading levels and digital proficiency levels, the chatbot is inclusive and simple to use for all beneficiaries.

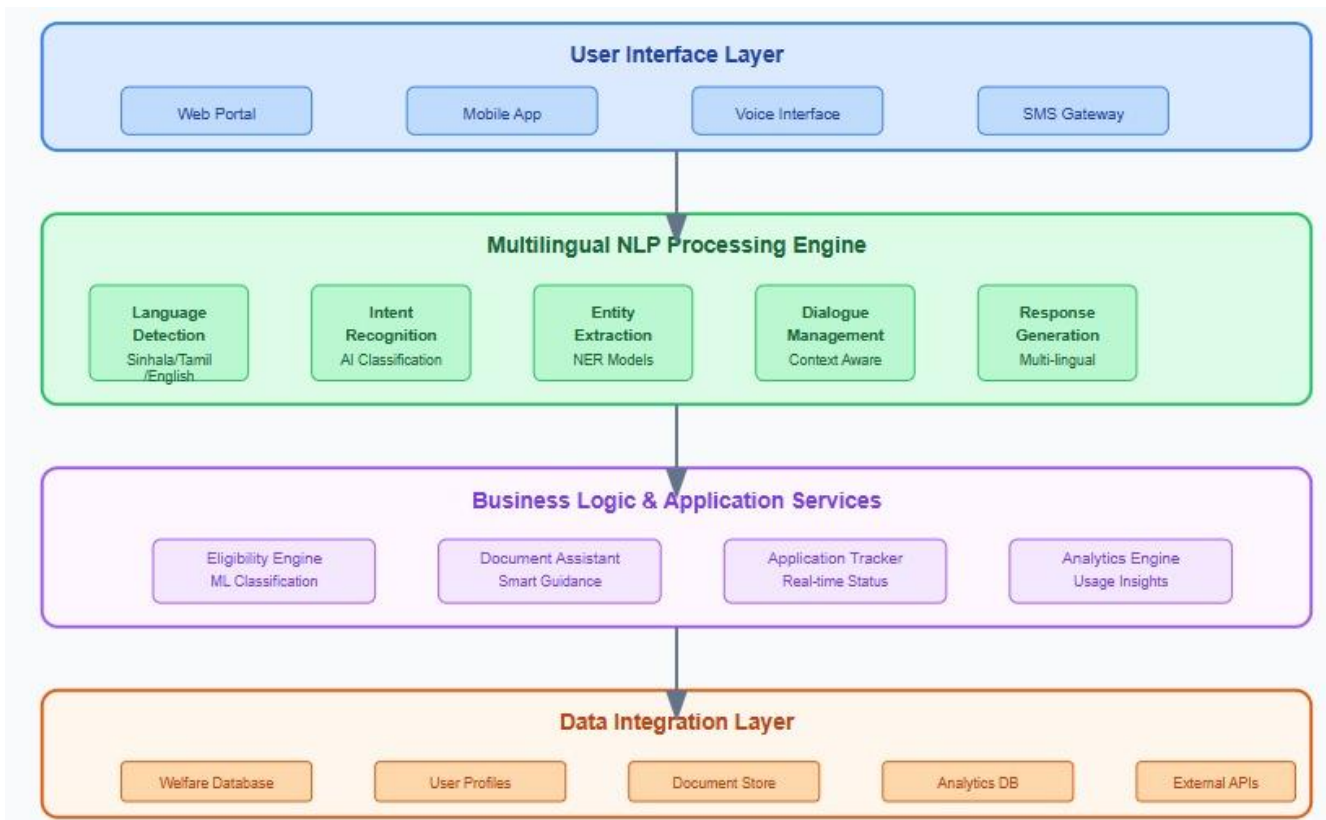


Figure 2: Multilingual Chatbot System Architecture Overview

The entire system architecture for the suggested multilingual chatbot driven by AI that aims to improve Sri Lanka's digital welfare services is shown in Figure 3. Scalability, maintainability, and effective handling of multilingual interactions in Sinhala, Tamil, and English are guaranteed by the architecture's tiered approach.

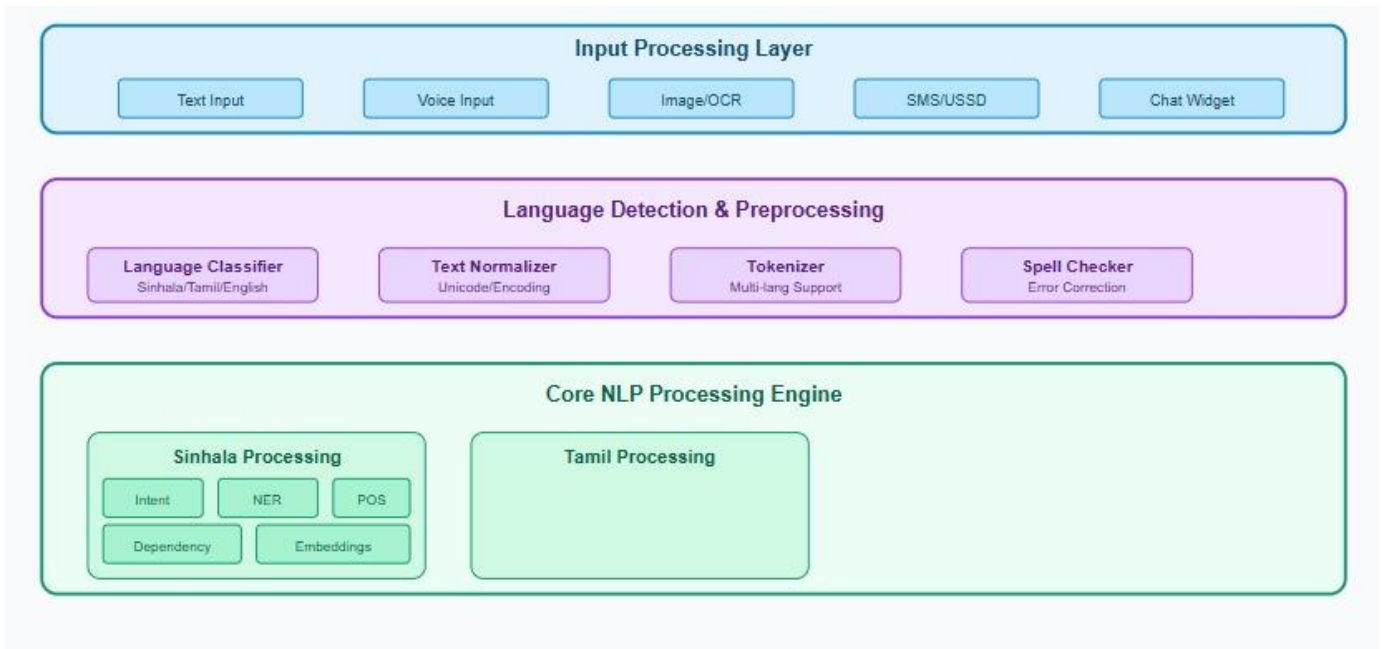
The main point of interaction is the User Interface Layer, which supports a variety of communication channels to suit a range of user preferences and technological capabilities. For users with reliable internet access, the web portal offers extensive functionality, and for smartphone users, the mobile application guarantees accessibility. By facilitating spoken interactions, voice interface integration overcomes literacy barriers and is especially advantageous for senior citizens and people with low reading proficiency. By reaching users with simple mobile phones, the SMS gateway makes sure that no citizen is left behind because of technological constraints.

The system's core intelligence is represented by the Multilingual NLP Processing Engine, which combines five interrelated components to orchestrate complex language understanding. By automatically determining whether users speak English, Tamil, or Sinhala, language detection directs requests to the proper processing modules. Whether a user is asking for application status, eligibility information, or document assistance, intent recognition uses machine learning classifiers trained on welfare-specific datasets to comprehend their goals. Important information components like application numbers, citizen identification information, or specific welfare program references are found through entity extraction. Natural, cohesive interactions are made possible by the dialogue management system, which preserves conversational context throughout several exchanges. Response generation produces linguistically correct and culturally relevant responses in the user's preferred language.

Welfare-specific features are implemented by the Business Logic and Application Services layer using specialized modules. To determine a citizen's eligibility for a given program, the eligibility engine uses machine learning classification. The document assistant offers astute advice on how to complete applications and necessary paperwork. While the analytics engine produces insights for policy improvement, the application tracker provides real-time status updates.

By establishing connections with current government databases, the fundamental Data Integration Layer guarantees the delivery of accurate and current information while upholding the security procedures and privacy compliance necessary for managing sensitive citizen welfare data.

11.1 Individual System Diagram



*Figure 3: Natural Language Processing (NLP) Engine
Component Diagram*

I am in charge of creating this component, which focuses on creating a multilingual chatbot system that improves the inclusion and accessibility of Sri Lanka's online welfare programs. By offering real-time assistance in Sinhala, Tamil, and English, the chatbot helps people communicate with welfare systems more effectively.

There are four subcomponents that comprise the system:

In order to guarantee inclusivity across urban and rural areas, the input processing layer manages a variety of citizen input methods, including text messages, voice input, SMS/USSD, OCR-based forms, and chat widgets.

Language detection and preprocessing: This includes text normalization, tokenization, spell checking, and a language classifier (English, Tamil, and Sinhala) to get the input ready for precise processing.

Natural language processing methods like intent recognition, Named Entity Recognition (NER), part-of-speech tagging, and embeddings are applied by the Core NLP Processing Engine. To guarantee accuracy, separate models are kept for Tamil and Sinhala.

Response Generation & Integration: Provides immediate assistance on eligibility checks, document submission, and application status by integrating the chatbot's responses with the officer dashboard, welfare application tracking system, and citizen site.

12. Gantt Chart

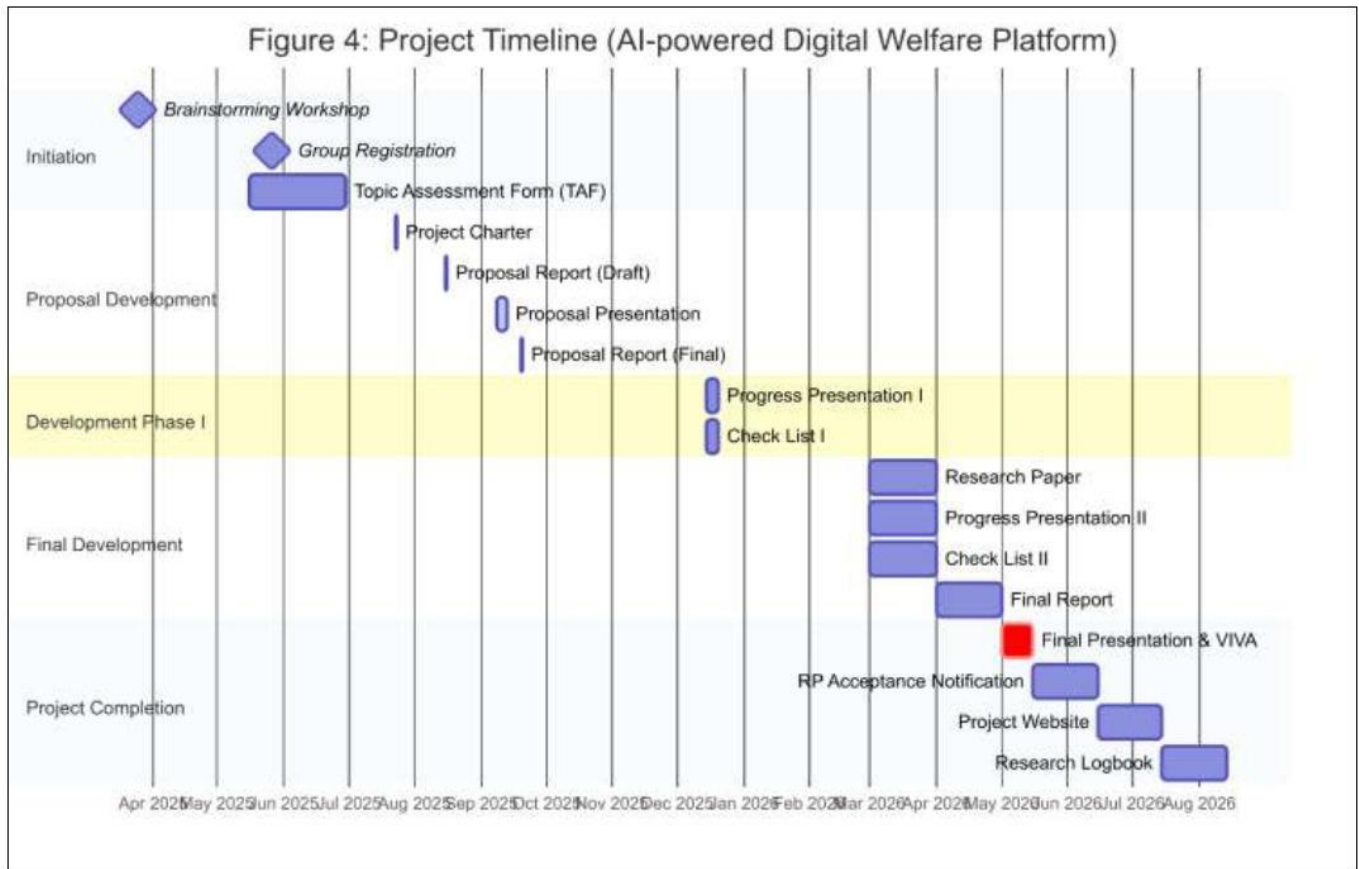


Figure 4: Gantt Chart

13. Project Requirements

13.1 Functional Requirements

In order for the multilingual chatbot system to efficiently serve welfare beneficiaries in Sinhala, Tamil, and English, it must perform specific actions and responsibilities that are outlined in the functional criteria.

- **Handling Multilingual User Queries:**
The chatbot will take customer inquiries in English, Tamil, and Sinhala via voice and text input. In order to comprehend user intent regarding welfare services like eligibility questions, document guidance, and application tracking, it must precisely detect the language and process inputs.
- **Identification of Intent and Extraction of Entities:**
In order to facilitate accurate query comprehension and response creation, the system will analyze user inputs to determine intentions (such as application status check, document submission assistance) and extract pertinent items (such as application number, document type).
- **Handling Conversations:**
The chatbot will manage dialogue flows based on user responses, provide clarifications when necessary, handle unexpected or confusing inputs using fallback mechanisms, and retain conversational context over several turns.
- **Integration of Welfare Data in Real Time:**
In order to provide dynamic and precise responses, the chatbot needs to establish connections with backend welfare databases and APIs to retrieve current data on user applications, eligibility conditions, and document requirements.
- **Context-Aware and Tailored Reactions:**
The generated responses ought to be linguistically correct in the user's native tongue, culturally sensitive, and contextually acceptable. Both templated and dynamically created answers based on real-time data should be supported by the system.
- **Support for Multiple Platforms:**
The chatbot should be available on both web and mobile platforms, facilitating communication through user-friendly chat windows or applications that are usable by people with different degrees of digital literacy.
- **Logging User Interactions and Gathering Feedback:**
All interactions will be recorded by the system for performance tracking, and users will be able to comment on chatbot responses for ongoing enhancement.

13.2 Non-Functional Requirements

Non-functional requirements describe the qualities of the system that make it work well, rather than the specific features it offers. For the multilingual chatbot designed to improve welfare service delivery, these requirements are:

- **Security:**
The system needs to keep users' personal and related information private and safe.
It should use strong rules for who can access data, protect information while it's being sent or stored, and follow local laws about data protection to make sure users' information isn't accessed by unauthorized people.
- **Reliability:**
The chatbot should always be available and quick to respond, offering support without any breaks.
It must manage user requests quickly in multiple languages and keep running smoothly even when many people are using it at the same time, so that everyone who needs help can get it without interruption.
- **Efficiency:**
The system should process user questions in different languages, understand what the user is asking, and quickly retrieve information from the database to give accurate and timely answers.
It should use resources wisely to ensure fast responses on both web and mobile devices, even when internet speed or device power is limited.
- **Usability:**
The chatbot's interface and way users interact with it must be easy to use, especially for people who may not be very familiar with technology or who speak different languages like Sinhala, Tamil, or English.
It should have clear instructions, simple language, and allow users to input information through text or voice to fit different needs.
- **Scalability:**
The system must be built to handle more users and more data over time without slowing down.
It should be able to grow both by adding more powerful computers and by expanding the number of computers used, so it can serve all welfare beneficiaries across the country as more people use it.

14. Commercialization

14.1 Budget

An approximate cost breakdown for the implementation and upkeep of the suggested AI-powered multilingual chatbot for Sri Lanka's digital welfare services is shown in the budget table that follows. The prices are estimates that could change based on the system size, chosen service providers, and vendor pricing.

Table 2: Budget Breakdown

Description	Cost	Occurring
Server Hosting(Cloud VPS)	LKR 10,000	Monthly
App Store Hosting(iOS Developer Program	LKR 7,500	Annual
Play Store Hosting (Google Play Developer)	LKR 30,700	One-time
Cloud Database Storage (Amazon S3 / Firebase)	0.023 USD per GB	Monthly
Domain Registration & SSL Certificate	LKR 5,000	Annual
Third-Party API Costs (Health Data, AI Models)	LKR 15,000	Monthly
Marketing & Promotion	LKR 50,000	Initial Launch
Maintenance & Technical Support	LKR 25,000	Monthly

14.2 Commercialization Methods

In order to increase accessibility and efficiency in social welfare services, the suggested AI-powered Multilingual Chatbot will be presented as a scalable digital welfare technology solution that can be embraced by both governmental and nonprofit organizations. B2B alliances and direct-to-consumer (D2C) business models are the main emphasis of our commercialization strategy:

1. B2B Partnership:

Work together with welfare offices, government agencies, and non-governmental organizations to incorporate the chatbot into their systems for providing services.

Provide institutions with tiered pricing based on user volume and access to premium features through subscription-based licensing.

2. Direct-to-Consumer (D2C)

Give residents access to a web platform and mobile application so they may communicate with the chatbot in English, Tamil, and Sinhala.

Take a freemium approach:

Free Tier: Basic document assistance, application monitoring, and access to general welfare data.

Premium Tier: Integration with personal welfare records, voice-based communication, priority responses, and advanced multilingual support.

For people and organizations that need more sophisticated functionality, monthly subscription levels will be offered.

3. Market Expansion

Targeting both rural and urban communities that encounter accessibility and language barriers while dealing with welfare programs, the inaugural launch will take place in Sri Lanka.

Extend regionally to South Asian nations with comparable social welfare management challenges and linguistic diversity.

4. Revenue Streams

subscription costs (both institutional and individual).

High-end multilingual support services (voice chat, sophisticated natural language processing).

granting third-party government and welfare platforms licenses to use the chatbot framework.

insights from aggregated and anonymised data for study and policymaking (in accordance with privacy standards)

15. References

- [1] M. U. Ahmed, "Digital transformation of social welfare services: Challenges and opportunities in South Asia," *South Asian Journal of Policy and Governance*, 2019.
- [2] United Nations, *E-Government Survey 2022: Digital government in the decade of action for sustainable development*. New York, NY, USA: United Nations, 2022.
- [3] ICTA Sri Lanka, *Digital Government Strategy 2023*. Colombo, Sri Lanka: ICTA, 2023.
- [4] S. Bhatnagar, *E-Government: From Vision to Implementation*. SAGE Publications, 2004.
- [5] S. B. Kotsiantis, "Supervised Machine Learning: A Review of Classification Techniques," *Informatica*, vol. 31, no. 3, pp. 249–268, 2007.
- [6] E. Adamopoulou and L. Moussiades, "Chatbots: History, Technology, and Applications," *Machine Learning with Applications*, vol. 2, p. 100006, 2020.
- [7] M. F. Goodchild, "Citizens as Sensors: The World of Volunteered Geography," *GeoJournal*, vol. 69, pp. 211–221, 2007.
- [8] World Bank, *Bridging the Digital Divide in South Asia*. Washington, DC, USA: World Bank, 2019.
- [9] Information Commissioner's Office, *Data Protection and Privacy*. Wilmslow, UK: ICO, 2021.
- [10] R. Kumar and M. L. Best, "Impact and Sustainability of E-Government Services in Developing Countries: Lessons Learned from Tamil Nadu, India," *Information Technology for Development*, vol. 12, no. 4, pp. 305–310, 2006.